Biochemistry Cume

JANUARY 25, 2002

Membranes

1. (20 points) Two new proteins with identical sizes and amino acid compositions are isolated, one of which is a soluble protein and the other a membrane protein. Discuss specific methods for analyzing the amino acid sequences that would lead to a prediction as to which protein will be the membrane protein.

2. (20 points) Define these terms:

    a. Leader sequence.

    b. Signal peptidase.

3. (60 points) Each method below has been used to gain information about which parts of a membrane protein are buried in a lipid bilayer, and which parts are exposed to aqueous solution. Choose any three of the methods below. For each choice, discuss in detail a complete experimental strategy for using the method to learn about the topology of a membrane protein. Be specific in your answers. ("Hand waving" responses will not be rewarded.)

    a. Fluorescence spectroscopy.

    b. Protease digestion.

    c. Mass spectroscopy.

    d. ESR spectroscopy.

    e. Chemical modification.